

Software Engineering Lab-1

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Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall record vehicle entry time via license plate scan and compute the total billing amount upon exit based on tiered hourly duration.	High	Pass: Correct entry time is recorded and the fee matches the applicable tier for the elapsed parking duration. Fail: Incorrect entry time or billing tier is recorded.	Ensures accurate parking duration and billing.
FR-002	Functional	The system shall allow a vehicle owner to reserve an available parking bay for a specified time period.	High	Pass: When an available bay and valid time period are selected, the system creates a reservation and marks the bay as reserved. Fail: Reservation is created for an unavailable bay.	Core functionality for smart parking space rental.
FR-003	Functional	The system shall validate a vehicle's license plate against its active parking reservation at the entry gate.	High	Pass: A vehicle with a matching active reservation is granted entry. Fail: A vehicle with no matching or expired reservation is denied entry.	Prevents unauthorized vehicles from entering reserved parking spaces.
FR-004	Functional	The system shall display the current parking status of each bay as available, reserved, or occupied.	Medium	Pass: The displayed status matches the actual recorded state of the parking bay after each reservation, entry, or exit event. Fail: The displayed status differs from the recorded state.	Helps vehicle owners and wardens identify available and occupied spaces.
FR-005	Functional	The system shall flag a vehicle when its recorded parking duration exceeds the permitted reservation period.	High	Pass: A vehicle exceeding its reservation period is flagged and the parking warden can view the alert. Fail: An overstay is not flagged.	Enables wardens to identify unauthorized overstays.

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NFR-001	Nonfunctional – Performance & Security	The gate camera license plate lookup API shall verify parking reservation validity in under 300 ms.	High	Pass: Benchmarking confirms reservation validation completes in less than 300 ms under simulated peak load. Fail: Validation exceeds 300 ms.	Ensures fast and responsive gate access while maintaining secure reservation validation.
NFR-002	Nonfunctional – Reliability	The system shall maintain accurate parking bay and reservation status during normal operating conditions.	High	Pass: In at least 99 out of 100 simulated reservation, entry, and exit transactions, the stored bay and reservation status matches the actual transaction state. Fail: More than one transaction produces an inconsistent state.	Prevents double-booking and incorrect parking availability information.

1. Preconditions

The smart parking system is operational.

Parking bay availability information is available.

The Vehicle Owner has provided a vehicle license plate.

A parking bay is available for the requested time period.

2. Postconditions

The selected parking bay is reserved.

The reservation is associated with the vehicle license plate.

The reservation time period is stored.

The parking bay status is updated to Reserved.

3. Main Success Scenario

Vehicle Owner selects “Reserve Parking Bay”.

System displays available parking bays.

Vehicle Owner selects an available parking bay.

Vehicle Owner enters the vehicle license plate and desired parking duration.

System verifies that the selected bay is available.

System validates the reservation details.

System creates the parking reservation.

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System updates the selected bay status to Reserved.

System displays the reservation confirmation.

Use case ends successfully.

4. Alternate Flow — Selected Parking Bay Becomes Unavailable

The system detects that the selected parking bay is no longer available.

System informs the Vehicle Owner that the selected bay is unavailable.

System displays other available parking bays.

Vehicle Owner selects another available bay.

System resumes from Step 4 of the Main Success Scenario.

